Calculus II Lesson 26 Conics Sections - Ellipse

Ellipse Elongated Horizontally

Equation: $\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$; $a>b$

$a^2=b^2+c^2$ Center of Ellipse: (h,k)

Vertices: $(h\pm a,k)$ Foci: $(h\pm c,k)$

Ellipse Elongated Vertically

Equation: $\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1$; $a>b$

$a^2=b^2+c^2$ Center of Ellipse: (h,k)

Vertices: $(h,k\pm a)$ Foci: $(h,k\pm c)$

1) $\frac{x^2}{4}+\frac{(y+2)^2}{25}=1$

a) Find the center of the ellipse:_____

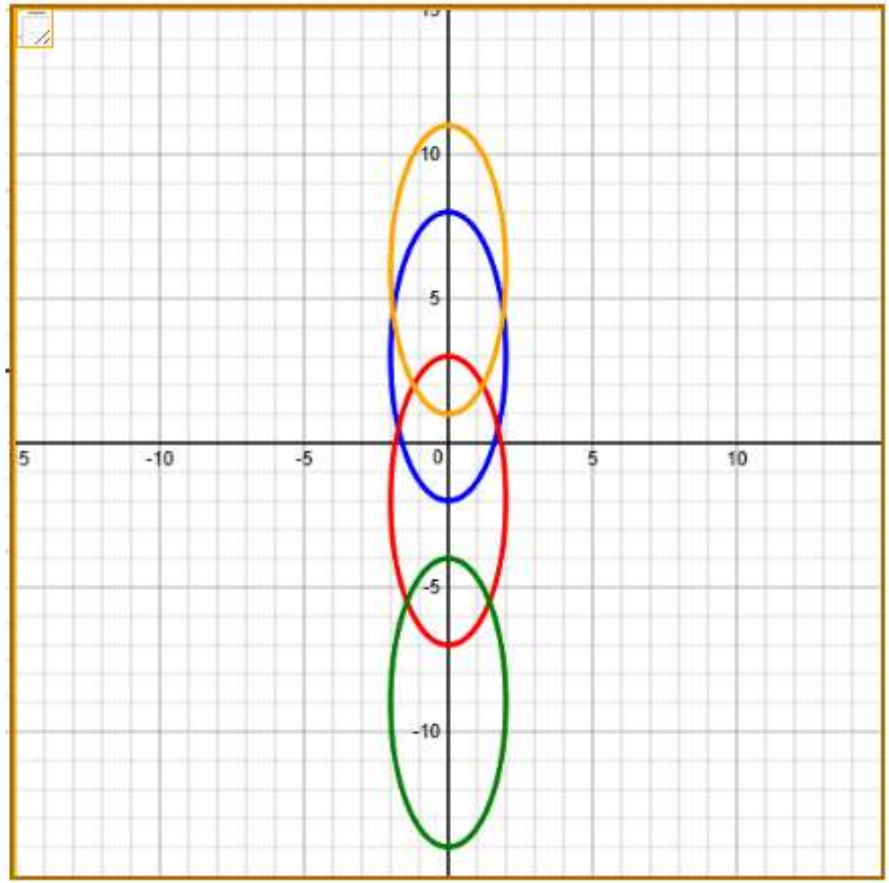
b) Find the vertices:_____

c) Find the foci:_____

d) Graph $\frac{x^2}{4}+\frac{(y+2)^2}{25}=1$

(multiple choice).

Choose the correct graph (blue, red, green, or orange)



Answers (#1)

a) b) c) d)

2) $\frac{(x-1)^2}{25}+\frac{(y+2)^2}{4}=1$

a) Find the center of the ellipse:_____

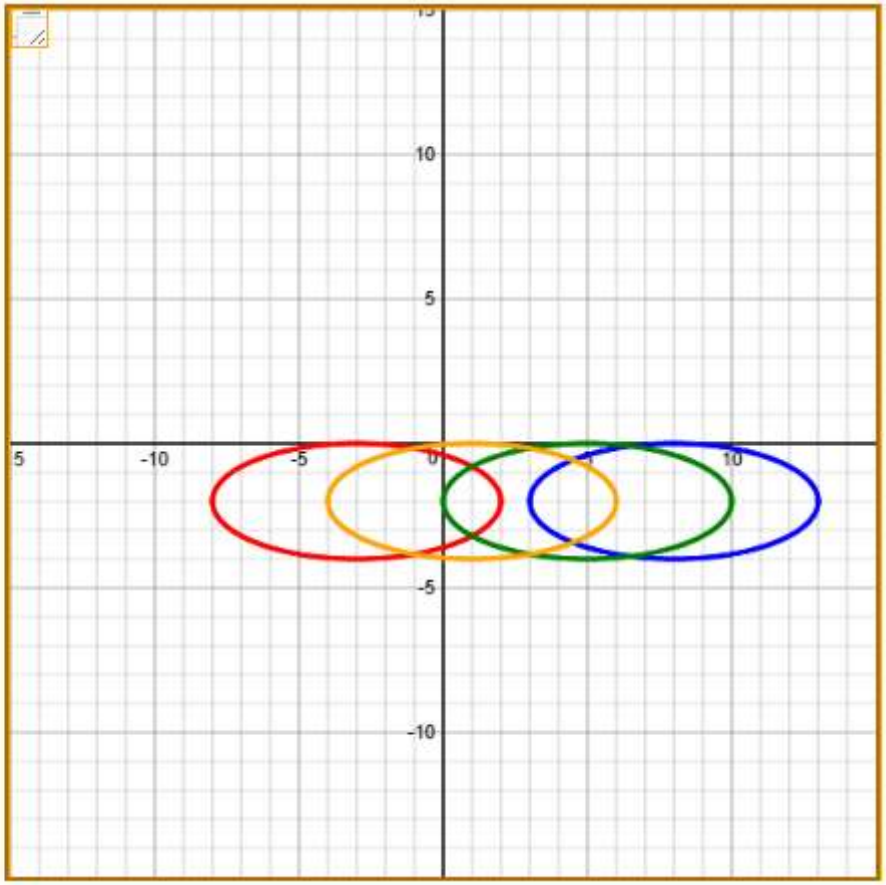
b) Find the vertices:_____

c) Find the foci:_____

d) Graph $\frac{(x-1)^2}{25}+\frac{(y+2)^2}{4}=1$

(multiple choice).

Choose the correct graph (blue, red, green, or orange)



Answers (#2)

a) b) c) d)

3) $4x^2 + 9y^2 + 24x - 36y + 12 = 0$

a) Find the center of the ellipse:_____

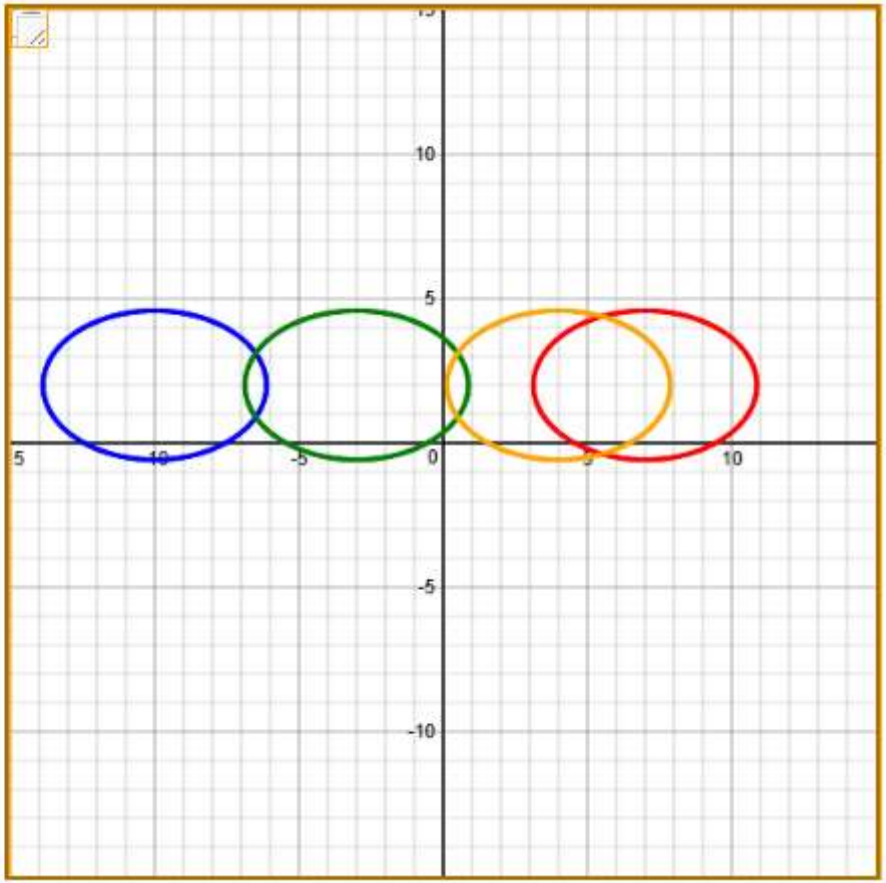
b) Find the vertices:_____

c) Find the foci:_____

d) Graph $4x^2 + 9y^2 + 24x - 36y + 12 = 0$

(multiple choice).

Choose the correct graph (blue, red, green, or orange)



Answers (#3)

a) b) c) d)

4) Find an equation of the ellipse that has the following:

Center:(0, 0) Vertex: (5, 0) ; Focus(4, 0)

a) a = distance from center to vertex = _____

b) c = distance from center to focus = _____

c) $b^2 =$ _____ Note: For ellipse, $b^2 + c^2 = a^2$

d) Equation of ellipse: _____

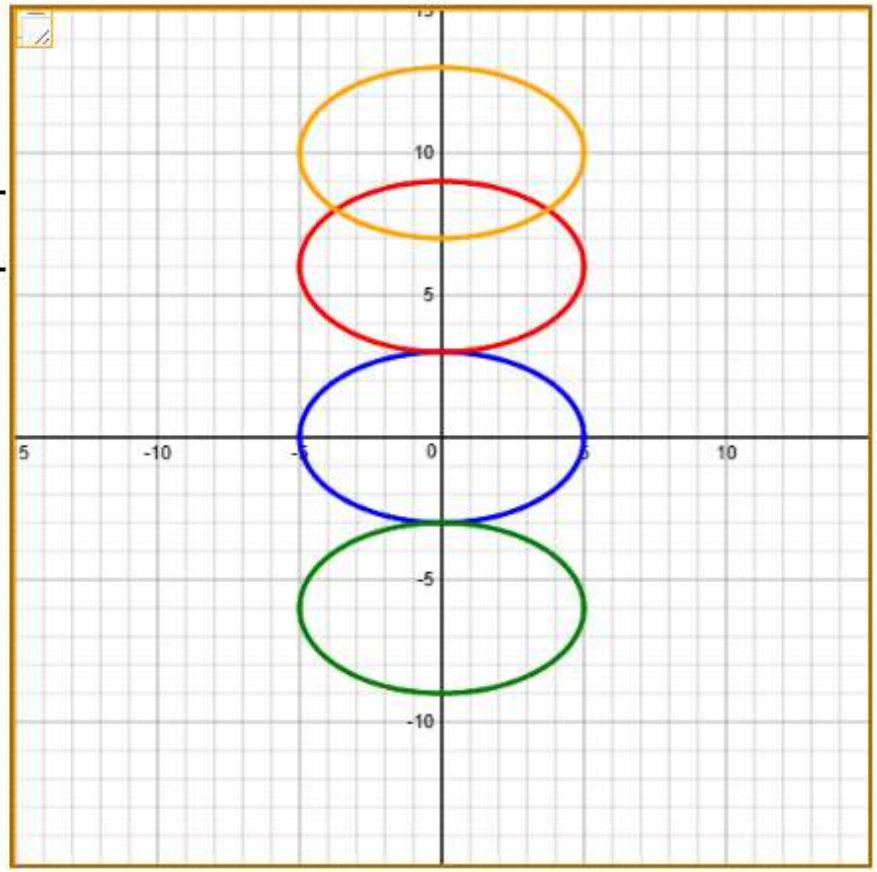
A) $\frac{x^2}{9} + \frac{y^2}{25} = 1$ B) $\frac{x^2}{25} + \frac{y^2}{9} = 1$

C) $\frac{x^2}{5} + \frac{y^2}{3} = 1$ D) $\frac{x^2}{3} + \frac{y^2}{5} = 1$

e) Graph equation of ellipse.

(multiple choice).

Choose the correct graph (blue, red, green, or orange)



Answers (#4)

a) b) c) d) e)

5) Find an equation of the ellipse that has the following:

Vertices: (8, 0) and (2, 0); Eccentricity: $\frac{3}{4}$ $\quad e = \frac{c}{a}$

a) a = distance from center to vertex = _____

b) c = distance from center to focus = _____

c) b^2 = _____ Note: For ellipse, $b^2 + c^2 = a^2$

d) Equation of ellipse:_____

Answers (#5)

a) b) c) d)

6) Find an equation of the ellipse that has the following:

Center:(1, 2) Vertex: (7, 2) ; Focus(5, 2)

a) a = distance from center to vertex = _____

b) c = distance from center to focus = _____

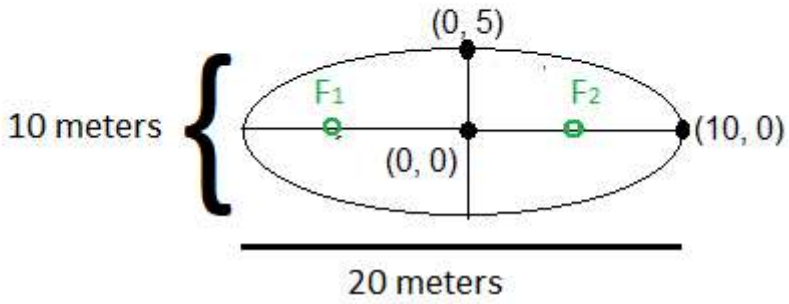
c) $b^2 =$ _____ Note: For ellipse, $b^2 + c^2 = a^2$

d) Equation of ellipse:_____

Answers (#6)

a) b) c) d)

7) A pond was built in shape of an ellipse. The pond is 20 meters long and 10 meters across. Suppose we want to place fountains at the foci. How far from the center of the pond should we place the fountains.



A) a = _____ (distance from center to vertex)

B) b = _____

C) c = _____ (distance from center to focus)

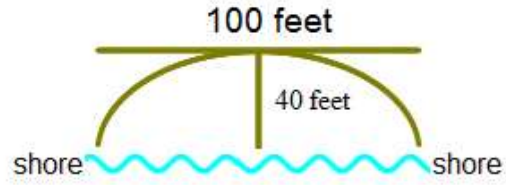
D) Where do we place the fountains?

Answers (#7)

A) B) C) D)

8) A bridge is designed in the shape of an elliptical arch.

The road over the bridge is 100 feet long and the maximum height is 40 feet.

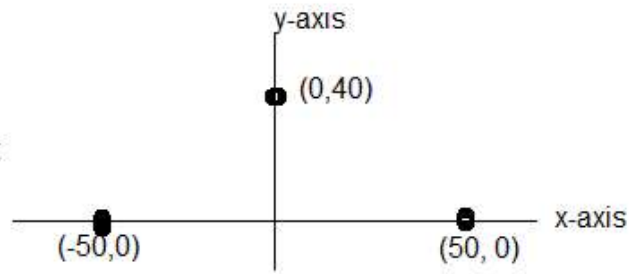


A) Write an equation for the elliptical arch..

Hint: Center of ellipse is (h, k) = (0, 0).
Distance from center to vertex = 50

B) When x = 30, y = _____

C) Can a barge 60 feet wide, with its highest point 34 feet above the water level, pass beneath the bridge?



Answers (#8)

A) B) C)